

Progress Report

- Increment 2 -

Group #1

Please use this template to describe your progress on the group project in the latest increment. Please do not change the font, font size, margins or line spacing. All the text in italic should be removed from your final submission.

1) Team Members

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2) Project Title and Description

Shiftsync

An employee scheduler application that allows managers to automate scheduling processes for their business. Users can log in as an employee or admin/manager. Employees can add their schedule and request to swap shifts. A manager/admin can add/remove employees, generate a master schedule accommodating to everyone's availability. Both can have a calendar view of their schedule.

3) Accomplishments and overall project status during this increment

During this increment, we expanded the core functionality by implementing both employee and admin features. We developed a complete login/logout system with validation and error handling, along with secure password functionality including hashed passwords and a limit of three login attempts for employees. For admins, we implemented a separate authentication flow with the ability to set and reset passwords without attempt restrictions.

Additionally, we enhanced the employee dashboard by adding key features such as shift swap requests, including the ability to submit, approve, and deny requests. We also implemented full availability management, allowing users to add, edit, and delete their availability, and improved the dashboard to display upcoming shifts and request status. The schedule logic was further developed to support these features.

Overall, the project is progressing well and aligns with the original scope. While the current system supports shift swap requests and availability management, further logic for handling overlapping shifts and expanding swap request to all employees will be addressed in the next increment. We now have a strong foundation with authentication, scheduling and user interaction features in place, setting up the project well for further expansion and refinement.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

One of the main challenges during this increment was implementing secure password handling, particularly ensuring proper hashing and validation across both employee and admin workflows. Debugging login issues, such as incorrect password errors and database inconsistencies, required careful testing and verification.

Another challenge involved coordinating work across multiple team members using Git. There were instances where changes were pushed to incorrect branches or merged incorrectly, which caused some confusion when linking issues and pull requests. These issues were resolved by improving branch management and following a more structured workflow.

In terms of project scope, we expanded the original plan by adding more advanced functionality to the employee dashboard, including shift swapping and availability management, which were not initially as detailed. While this increased complexity, it significantly improved the useability and functionality of our application.

Overall, despite some technical and workflow challenges, our team was able to resolve these issues effectively and continue to make strong progress during this increment.

Team Member Contribution for this increment

Jaliah:

- a) Progress Report:
Contributed to documenting the implementation of the admin login/logout system, including credential validation and UI styling updates.
- b) Requirements and design document:
Helped define requirements for the admin authentication system, including default admin credentials and validation checks for login errors.
- c) Implementation and testing document:
Documented testing for admin login functionality, including successful login cases and error handling.
- d) Source:
Contributed to the admin authentication system by creating and modifying the following files:
 - admin.html- created the admin login
 - login.html- updated login flow to support admin authentication
 - initadmin.py- initialized admin credentials in the database
 - init_db.py- updated database schema to include admin-related data
 - main.py- added admin login/logout
- e) Video:
Nikki did the video.

Isabella

- a) Progress Report:
Documented the implementation of password security features, including login attempt limits and admin password management.
- b) Requirements and design document:
Defined requirements for limiting users to 3 login attempts, admin password system without attempt restrictions, and ability for admin to reset/change passwords.
- c) Implementation and testing document:
Documented login attempt restriction testing, password reset workflow, and admin password validation and hashing.
- d) Source:

Enhances password and authentication system by contributing to:

- adminpassword.html- created admin password input interface
- admin_reset_password.html- implemented admin password reset page
- main.py- added: login attempt tracking (max 3 attempts, password reset logic, admin password handling (no attempt restriction), password hashing and validation
- ShiftSyncDB.db- updated to store hashed passwords and track login attempts

e) Video:
Nikki did the video.

Annika

- f) Progress Report:
Wrote the section describing enhancements to the employee dashboard and shift swap functionality.
- g) Requirements and design document:
Defined requirements for shift swap request system, approval/denial workflow, availability management, and dashboard display improvements.
- h) Implementation and testing document:
Documented swap request submission and approval flow, availability add/edit/delete functionality, and dashboard updates displaying shifts and requests
- i) Source:
Expanded dashboard and scheduling features:
- dashboard.html- updated dashboard UI to display upcoming shifts, swap request statuses, availability summary
 - swap_requests.html- implemented UI for submitting and viewing shift swap requests
 - availability.html- enhanced availability input functionality
 - availability_edit.html- added ability to edit availability
 - insertinto.py- updated logic to support swap requests and availability data
 - main.py- swap request submission, approve/deny logic, shift transfer updates, availability add/edit/delete functionality
- j) Video:
Nikki did the video.

Nikki

- k) Progress Report:
Contributed to documenting scheduling logic enhancements, system integration, and user interface improvements across the application.
- l) Requirements and design document:
Define scheduling-related requirements, including how shifts interact with availability and swap requests.
- m) Implementation and testing document:
Documented testing and validation of shift swapping functionality, including identifying and fixing errors related to swap request processing and system updates.
- n) Source:
Continued development of scheduling logic by contributing to:

- main.py- fixed errors related to shift swapping logic, improved integration between swap requests, scheduling, and availability features
- templates/- added and refined CSS styling across application pages

o) Video:
Completed the project video.

5)Plans for the next increment

For the next increment, we plan to enhance both the scheduling logic and administrative functionality. One major focus will be improving the Shiftswap system by allowing employees to submit open swap requests without specifying a particular employee. This will enable other employees to view and accept available swap requests more flexibly. Additionally, the system will be updated to handle overlapping shift requests, ensuring that conflicts are properly detected and resolved within the scheduling logic. We also plan to further develop the admin dashboard by adding management features. These enhancements will allow administrators to view and edit employee schedules, manage employee availability, and monitor control shift assignment more effectively. In addition, improvements will be made to the integrations between scheduling, availability, and swap requests to ensure smoother system behavior and more accurate updates across all features. Overall, the next increment will focus on strengthening the system's core scheduling capabilities while expanding administrative control and usability.

6)Stakeholder Communication

Dear Stakeholders,

We hope you are doing well. We wanted to provide you with an update on the current progress of the ShiftSync scheduling system.

Over the past development phase, our team has made significant progress in improving both user experience and system functionality. We have successfully implemented both user experience and system functionality. We have successfully implemented a secure login system for both employees and administrators, including enhanced password protection measures to ensure account security. Additionally, we expanded the employee dashboard to provide a clearer view of schedules, availability, and shift-related activity.

A key improvement in this phase was the introduction of a shift request and swap feature. Employees can now submit requests and review incoming requests directly through the system, creating a more flexible and efficient scheduling process. We also improved how availability is managed allowing users to update their preferences more easily.

As with any growing system, we encountered some challenges related to integrating new features with existing scheduling logic. In particular, ensuring that shift changes and requests update accurately across the system requires additional refinement. These types of integration challenges are common when scaling functionality, and we have been actively addressing them to improve overall reliability.

Looking ahead, our next phase will focus on strengthening scheduling capabilities and administrative tools. We plan to enhance how shift conflicts are handled, expand administrative controls for managing schedules and availability, and continue improving the overall usability of the platform.

We appreciate your continued support and look forward to share further updates as our development progresses.

Sincerely,
The ShiftSync Team

7)Link to video

<https://drive.google.com/file/d/1rC88MWpUuutwwRCZAEtU2iy-KYXetkPc/view?usp=sharing>